Shweta Choudhary

Prime Minister's Research Fellow

Personal Profile



Gender	Female
Nationality	Indian
Address	Department of Chemistry, Indian Institute of Technology Roorkee, 247667, Haridwar, Uttarakhand, India
Email	shweta_c@cy.iitr.ac.in
Area of research	Computational chemistry

Education

Dec 2022 – present **PhD in Computational Chemistry**, *Department of Chemistry, Indian Institute of Technology Roorkee*, India

2020–2022 **Master of Science in Chemistry**, *Department of Chemistry, Indian Institute of Technology Ropar*, India, CGPA: 8.88/10

2017–2020 **Bachelor of Science (Honours) in Chemistry**, *Department of Chemistry, Gargi College, University of Delhi*, India, CGPA: 9.23/10

Research Experience

2021–2022 **Master's thesis**, *"First-principle investigation for low-dimensional anodes for Na-ion batteries"*, Indian Institute of Technology Ropar, India

Advisor: **Dr. T. J. Dhilip Kumar**

Publications

2026 Dixit, D., **Choudhary, S.**, Sarkar, S., Banerjee, S. & Boving, T. Experimental and computational investigation of selective removal of ciprofloxacin by HKUST-1. *Journal of Water Process Engineering* **86,** 109802 (2026)[DOI]

2026 Goyal, N., **Choudhary, S.**, Witulski, B., Yan, S., Singh, B., Kornienko, N., Banerjee, S. & Mathur, S. Selective Dinitrogen Electroreduction to Ammonia on Zn2V2O7 Pyrochlore. *ACS Catalysis* (2026)[DOI]

2025 **Choudhary, S.** & Banerjee, S. Ion coordination and migration mechanisms in alkali metal complex borohydride-based solid electrolytes. *Communications Chemistry* **8,** 123 (2025)[DOI]

2025 Saroha, R., **Choudhary, S.**, Brahma, R. & Banerjee, S. Designing superatomic (Li/Na/K)-Al-Sc cluster catalysts for selective electrochemical HER and NRR. *The Journal of Physical Chemistry C* **129,** 6562–6573 (2025)[DOI]

2024 **Choudhary, S.**, Saroha, R. & Banerjee, S. Efficient electron injection into graphullerene enables reversible NaC_2 sodium storage. *ACS Applied Materials & Interfaces* **16,** 50859–50869 (2024)[DOI]

2023 Saini, V.[†], Krishankant[†], **Choudhary, S.**[‡], Gaur, A., Banerjee, S., Bagchi, V. & Venkatesh, V. Atomically precise copper nanoclusters as a potential catalyst for the electrochemical oxygen evolution reaction. *Journal of Materials Chemistry A* **11,** 24754–24763 (2023) [‡]Second author[DOI]

2022 **Choudhary, S.**, Duhan, N. & Dhilip Kumar, T. J. Hydrogen passivated β_{12}-borophene nanoribbon: a propitious one-dimensional metallic anode for sodium-ion rechargeable batteries. *Applied Surface Science* **606,** 154825 (2022) [DOI]

Awards, Fellowships & Recognition

2026 Wellness Secretary, Himalaya Bhawan, IIT Roorkee

2025 Mess Secretary, Himalaya Bhawan, IIT Roorkee

2024 Best Poster Award, ChemDay 2024, Department of Chemistry, IIT Roorkee

2024 Best Poster Award, CRSI-ACS Early Career Researchers' Symposium 2024, KIIT Bhubaneswar

2023 Prime Minister's Research Fellowship (PMRF)

2022 Qualified the Graduate Aptitude Test in Engineering (GATE)

2020 Qualified the IIT Joint Admission Test for M.Sc. (IIT JAM)

Conferences, Workshops, & Presentations

Mar 2026 **Global Virtual Symposium: Computational Materials Discovery**, ACS Spring 2026. (Oral talk) [Certificate]

Jan 2025 **Evolution of Electronic Structure Theory & Experimental Realization (EESTER)**, SRM Institute of Science and Technology & IIT Madras. (Attended workshop & presented poster) [Certificate]

Dec 2024 **Conference on Advances in Chemistry for Energy and Environment (CACEE)**, TIFR Mumbai. (Poster presentation) [Certificate]

Nov 2024 **Research Scholar's Day (ChemDay)**, Department of Chemistry, IIT Roorkee. (Poster presentation) [Certificate]

Oct 2024 **CRSI-ACS Early Career Researchers' Symposium**, KIIT Bhubaneswar. (Poster presentation) [Certificate]

Jan 2024 **International Conference on Functional Materials**, IIT Kharagpur. (Poster presentation) [Certificate]

Experience with Theoretical Models, Methods & Packages

Models & methods	Density functional theory (DFT), harmonic and quasi-harmonic approximations, density functional perturbation theory (DFPT), nudged elastic band (NEB) method, growing string method, and classical and *ab initio* molecular dynamics simulations
Simulation software	VASP, FHI-aims, FHI-vibes, i-PI, Gaussian, Quantum ESPRESSO, CP2K, GROMACS, Critic2, and Materials Studio
GUI & Python modules	pymatgen, ASE, Phonopy, VASPKIT, VTST Tools, sumo, pyband, Aimstools, aimsChain, baderVis, VESTA, VMD, GIMS, GaussView, and Avogadro
Technical skills	Python (Anaconda), Bash/Shell scripting, LaTeX, High-performance computing (SLURM and PBS job schedulers)
Operating systems	Linux, macOS, Windows

Referees

Dr. Swastika Banerjee
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